22IT811 PROJECT WORK

Connect-X
AI Driven Department Collaborative Hub

A PROJECT REPORT

Submitted by

Shai Kumar R	**-**	**111722203099**
Teja Krishna C	**-**	**111722203112**

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BONAFIDE CERTIFICATE

Certified that this project report **Connect-X AI Driven Department Collaborative Hub** is the Bonafide work of **Shai Kumar R (111722203099), Teja Krishna C (111722203112)** who carried out the project work under my supervision**.**

SIGNATURE
Dr. M.Sheerin Banu, M.Tech., Ph.D.,
Professor and Head

Department of Information Technology
R.M.K. Engineering College
R.S.M. Nagar, Kavaraipettai,
Thiruvallur District–601206

SIGNATURE
Mr. N. Shanmugam M.E,
Supervisor
Assistant Professor
Department of Information Technology
R.M.K. Engineering College
R.S.M. Nagar, Kavaraipettai,
Thiruvallur District–601206

Submitted for the Project Viva-Voce held on ……………………………………
at **R.M.K. Engineering College**, Kavaraipettai, Thiruvallur District-601206.

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Connect-X: AI-Driven Department Collaborative Hub is a unified academic collaboration platform designed to streamline idea sharing, project formation, resource discovery, and mentorship across departments. Traditional departmental coordination relies on fragmented channels—emails, messaging apps, paper notices, and isolated portals—leading to missed opportunities, duplicated efforts, and slow decision cycles. Connect-X resolves these issues through an AI-assisted, privacy-aware web platform that centralizes communication, intelligently matches students and faculty to relevant projects or events, and provides real-time collaboration tools that fit typical academic workflows. At its core, Connect-X provides four capabilities. First, **AI-powered matchmaking** analyzes user profiles, skills, interests, and ongoing initiatives to recommend collaborators, mentors, or departmental resources such as labs and clubs. Second, a **secure idea workspace** supports structured proposal submission, versioning, and role-based visibility to protect early-stage intellectual work while enabling peer and mentor feedback. Third, a **contextual recommendation engine** surfaces calls for participation, seminars, funding notices, and relevant reference materials at the right time, reducing information overload. Finally, **event and community features**—discussion channels, announcements, and virtual meetups—create a living network that strengthens departmental culture and accelerates learning by doing. By consolidating scattered tools into a single, intelligent hub, Connect-X aims to improve discovery, inclusion, and throughput across the department. The result is a sustainable, data-informed collaboration ecosystem that reduces friction, promotes equitable access to mentorship and resources, and shortens the cycle from idea to validated project outcomes.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

S.NO.	ABBREVIATION	EXPANSION
1	AI	ARTIFICIAL INTELLIGENCE
2	ML	MACHINE LEARNING
3	DNN	DEEP NEURAL NETWORK

CHAPTER 1
INTRODUCTION

In the rapidly evolving landscape of the tech industry, technical interviews stand as critical gateways for individuals seeking to carve out successful careers. These assessments serve as pivotal moments where candidates demonstrate their proficiency, problem-solving abilities, and adaptability under pressure.

The aim of this project is to address the aforementioned challenges by developing specialized preparation tools tailored to the unique needs of individuals preparing for technical job interviews. While existing resources offer valuable insights, they often fall short in providing tailored guidance that accommodates the diverse skills and backgrounds of candidates. Moreover, traditional interviewing methods, with their mechanical formats, fail to capture the dynamic and uncertain nature of real technical interviews, leaving candidates ill-prepared for the interactive dynamics and evolving question structures they may encounter.

1.1 PROBLEM STATEMENT

Despite active student societies, mentoring cells, and research labs, departmental collaboration remains fragmented. Information about projects, events, and funding is dispersed across notice boards, emails, and social channels, making discovery unreliable. Students struggle to find teammates with complementary skills; faculty spend time re-circulating the same updates; and early ideas often stall due to lack of timely guidance. There is also limited support for secure, staged idea sharing—students either overshare in public channels or withhold details, both of which hinder feedback and progress. The absence of a unified, data-driven system results in duplicated efforts, uneven access to opportunities, and avoidable delays in proposal approvals and project onboarding. Connect-X addresses these pain points with a centralized, AI-assisted hub that enables structured discovery, secure collaboration, and mentor engagement, tailored to departmental needs.

1.2 LITERATURE SURVEY

Chou et al. [1] (2022) proposed an AI mock-interview platform for evaluation and analysis of interview performance. The system creates realistic interview scenarios, assesses candidate responses, and provides personalized feedback. It creates a low-stress environment for candidates to practice interviews and sharpen their skills. However, it proved difficult to authentically imitate the spontaneity of live interviews.

Nandhini et al. [2] (2021), the recommendation of an automated system for screening resumes has been made that takes out important information within a resume using natural language processing (NLP) and ranks candidates against the criterions of conformance with the given job. The study goes on to discuss the advantage of automation of the selection process and how it could enhance recruitment processes. However, problems with unstructured resumes and non-standard layouts serve as a major challenge in such an approach and, moreover, with limited flexibility toward multilingual documents.

1.3 SYSTEM REQUIREMENTS

1.3.1 HARDWARE REQUIREMENT

- Processor - i3, i5, i7
- RAM - 4 GB

1.3.2 SOFTWARE REQUIREMENTS

- Operating System - 10 (or) Later Windows Version
- Front End - Web Development Framework
- Back End - Java
- Database - MySQL

1.3.3 FEASIBILITY STUDY

All components are standard in modern web stacks; AI recommendations can begin with lightweight models (e.g., profile/keyword matching) and evolve to collaborative filtering as interaction data grows. Integration with institutional SSO or email is feasible via OAuth/LDAP. Storage needs are moderate and can be scaled.

- **Economical Feasibility** : Economic feasibility assesses whether the project can be implemented with reasonable financial investment. For Connect-X, the overall cost of development is moderate, as the system relies heavily on open-source technologies, which eliminates licensing or subscription expenses. The primary costs include hosting a virtual machine (VM), securing a domain name, implementing SSL certificates for data protection, and allocating time for developer resources.
- **Technical Feasibility :** Technical feasibility examines whether existing technical resources are sufficient to support system development. Connect-X is built using standard, widely supported web technologies. These technologies do not impose heavy hardware or software demands, making the system highly practical for departmental implementation.
- **Social Feasibility** : Social feasibility evaluates the level of acceptance and readiness among potential users—students, faculty, staff, and project coordinators. The goal of Connect-X is to complement existing workflows rather than replace them abruptly, which increases its chances of quick and positive adoption**.**

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Most academic departments currently rely on a **patchwork of tools**-emails for announcements, messaging apps for informal coordination, spreadsheets for tracking teams, and separate forms or PDFs for proposal submissions. Notices are duplicated across WhatsApp/Telegram groups, class representatives' messages, and physical notice boards. Because these channels are not integrated, information becomes **scattered and transient**. Students often miss deadlines or hear about opportunities late, while faculty and coordinators spend time reposting the same content and clarifying status updates individually.

For **idea sharing and team formation**, students typically approach peers within their own class or social circle. This narrows the pool of potential collaborators and leads to homogeneous teams that may lack complementary skills (e.g., design, documentation, data analysis, deployment). Mentorship usually begins only after a team is formed and a topic is tentatively chosen, which means **early feedback is lost** and many promising ideas never reach the proposal stage.

2.1.1 DISADVANTAGES OF EXISTING SYSTEM

- **Fragmented Communication:** Announcements and discussions span multiple channels with no single source of truth. Students must monitor several streams, increasing the chance of **missed information** and duplicated questions.
- **Inefficient Team Formation:** Without structured discovery, students rely on personal networks. This results in **skill-imbalanced teams**, slower formation, and less diversity of ideas.
- **Unstructured Proposal Workflow:** Email-based submissions cause **version conflicts**, unclear status, and inconsistent feedback quality. Deadlines and rubrics are not enforced by the tools themselves.
- **Limited Mentor Visibility:** Faculty cannot easily discover projects aligned with

their expertise or availability. This leads to **uneven mentor distribution** and delayed guidance during the idea-incubation stage.

- **Weak Data Security & Access Control:** Public groups are inappropriate for sharing early-stage ideas, while shared spreadsheets lack **granular permissions** and **audit trails**.
- **No Institutional Memory:** Useful conversations and decisions get buried in chats or personal mailboxes; there is no **central knowledge base** to learn from past cohorts, templates, or exemplar proposals.
- **Event & Community Gaps:** Registrations, attendance, and outcomes are not connected to user profiles. Departments cannot easily **analyze engagement** or identify students who might benefit from targeted opportunities.
- **Low Measurability:** Without dashboards, administrators lack **KPIs** (e.g., time from idea to approval, mentor load balancing, participation rates). This makes improvement reactive rather than data-driven.
- **Scalability and Sustainability Issues:** As batch sizes increase, manual coordination **does not scale**. Reliance on a few coordinators causes bottlenecks and burnout, risking delays near review weeks.
- **Inclusion Barriers:** Students who are new, shy, or outside dominant groups may be **under-represented** in projects due to lack of visibility and supportive on-ramps.

2.2 PROPOSED SYSTEM

Connect-X consolidates departmental collaboration into a **single, AI-assisted web platform** that aligns with academic milestones. The system's design centers on four pillars:

1. **Unified Communication & Discovery:** A central feed for announcements, opportunities, and events, with filters and tags mapped to departments, laboratories, and interest areas. Keyword search and topic tags ensure that relevant information is **easy to find and revisit**.
2. **Secure Idea Workspaces:** Students can draft ideas using structured templates (problem, objectives, scope, novelty). **Role-based access control (RBAC)** enables staged disclosure—private to team → mentor review → departmental review—preserving intellectual ownership while gathering feedback. Built-in versioning and inline comments eliminate email clutter.
3. **AI-Driven Matching:** A recommendation engine matches students to **teammates, mentors, labs, and events** based on skills, interests, prior coursework, and availability. The model starts with profile/keyword matching and can evolve into collaborative filtering as interaction data grows. This **broadens participation** beyond immediate social circles.
4. **Workflow Automation:** Proposal submission, review assignments, deadlines, rubrics, and approval statuses are managed within Connect-X. Automated reminders reduce follow-ups; dashboards show **where items are stuck**, allowing proactive intervention by coordinators.
5. **Events & Community:** Integrated registration, QR check-ins, session materials, and post-event reflections tie participation to profiles, enabling **evidence-based mentoring** (e.g., recommending a seminar to a student who is preparing for a related project).

6. **Analytics & Governance:** Coordinators and HoD can see aggregated metrics: mentor load, idea-to-approval cycle time, participation heatmaps, and module adoption. **Audit logs** preserve accountability and support compliance with departmental policies.
7. **Interoperability & Access:** Optional SSO via OAuth/LDAP, export of reports (CSV/PDF), and mobile-friendly UI ensure smooth adoption. Security uses TLS in transit and encryption at rest for sensitive data.

By embedding discovery, collaboration, and governance into one system, Connect-X **shortens time-to-project**, increases the quality of feedback, and provides institutional memory for continuous improvement.

2.2.1 ADVANTAGES OF PROPOSED SYSTEM

Single Source of Truth: Centralized announcements, proposals, and feedback eliminate duplication and confusion. Students and faculty know **exactly where to look** for updates.

Faster, Better Team Formation: AI-assisted matching connects complementary skill sets across sections and years, improving **team balance** and project outcomes.

Structured, Transparent Workflows: Proposal templates, rubrics, and status tracking create **clarity**. Everyone sees what is pending, by whom, and by when—reducing email back-and-forth.

Privacy & IP Respect: Role-based visibility and staged sharing protect early ideas while still enabling **timely mentorship** and peer input.

Mentor Load Balancing: Coordinators can visualize mentor availability and allocate projects fairly, avoiding **overburdening** a few faculty members.

Evidence-Based Decisions: Dashboards provide actionable KPIs (engagement, throughput, cycle times), enabling **data-driven** planning for events, electives, and resource allocation.

Higher Inclusion & Reach: Recommendation feeds surface relevant opportunities to students who might otherwise be **overlooked**, improving equity and participation.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

Fig 3.1 System Architecture of the Proposed System

The system architecture of **Connect-X: AI-Driven Department Collaborative Hub** is designed to ensure scalability, security, modularity, and smooth user experience. It follows a **multi-tier architecture** that separates the presentation layer, application logic, AI services, and data management components. This separation ensures maintainability, efficient data flow, and easy integration with institutional authentication systems.

3.2 UML DIAGRAMS

3.2.1 USE CASE DIAGRAM

A use case diagram is a type of behavioral diagram created from a Use-case analysis. The purpose of use case is

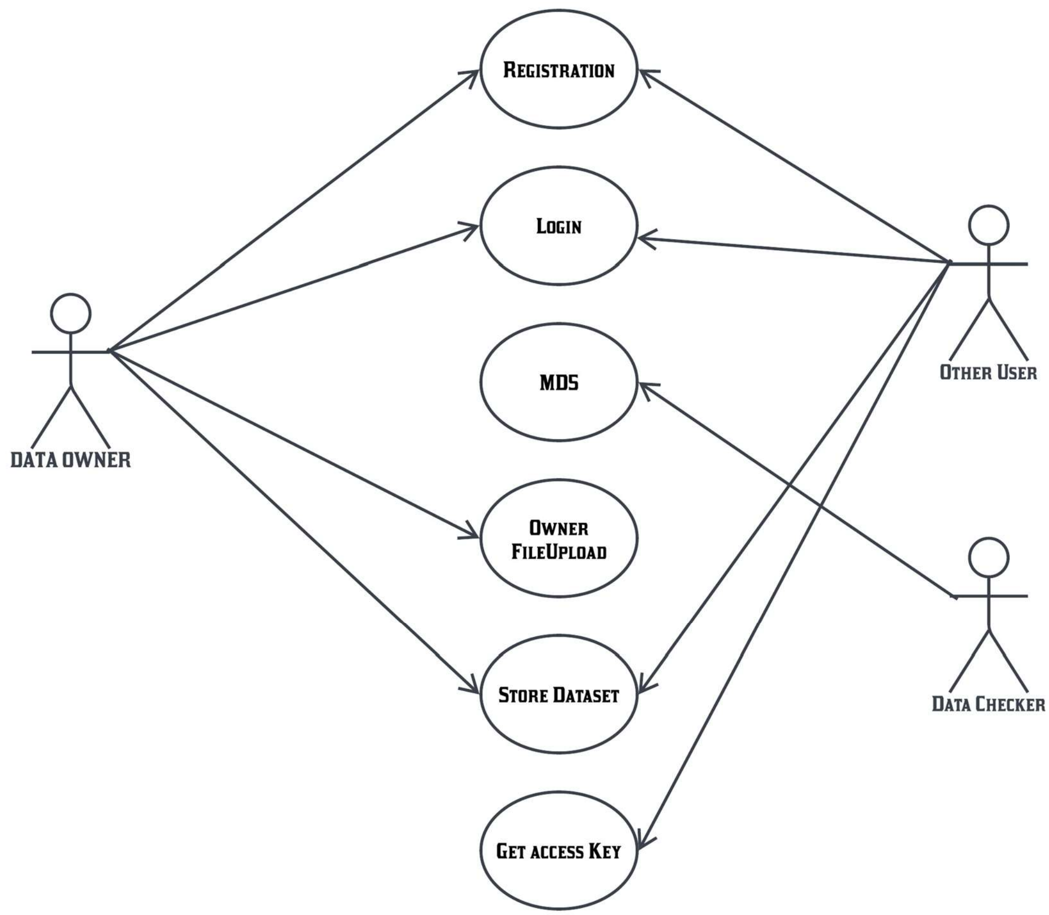

Fig 3.2.1 Use Case Diagram

3.2.2 CLASS DIAGRAM

The class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for

Fig 3.2.2 Class Diagram

3.2.3 ACTIVITY DIAGRAM

Activity diagram is basically a flow chart to represent the flow form one activity to another activity. The activity can be described as an operation of the system. So the control flow is drawn from one operation to another.

Fig 3.2.3 Activity Diagram

3.2.4 SEQUENCE DIAGRAM

Sequence diagrams are best suited to the portrayal of simple interactions among relatively small numbers of objects.

Fig 3.2.4 Sequence Diagram

CHAPTER 4
SYSTEM IMPLEMENTATION

4.1 MODULES

- USER AUTHENTICATION AND PROFILE MANAGEMENT
-
- FEEDBACK MECHANISM

4.2 MODULE DESCRIPTION

4.2.1 User Authentication and Profile Management

4.2.2 Adaptive Technical Interview Preparation

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4.5 TEST CASES

Table 4.5 Test Cases

TEST CASE ID	MODULE	TEST SCENARIO	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS

CHAPTER 5

RESULTS & DISCUSSION

The implementation of the AI-driven technical interview preparation project has yielded transformative outcomes, addressing the multifaceted challenges encountered by candidates in navigating technical interviews within the dynamic tech industry landscape. By harnessing advanced technologies such as natural language processing (NLP) and deep learning, the project has revolutionized

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

In conclusion, our project aims to

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APPENDIX I SOURCE CODE

APPENDIX II SCREENSHOTS

Fig. A.2.1 Login Page

Fig. A.2.2 Resume Upload Page